

SIMATS ENGINEERING
ASSESSMENT TOOL

CO2 - AS1

COURSE NAME: Theory of Computation

COURSE CODE: CSA1301

COURSE OUTCOME COVERED

CO1: Analyze fundamental mathematical concepts of automata theory for formal languages and decision problems.(BL:3)

Assessment Tool Weightage: 40%

QUESTIONS: 5
Duration : 1 Hour

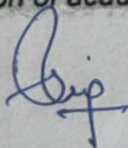
Total Marks:25

Analytical Problem

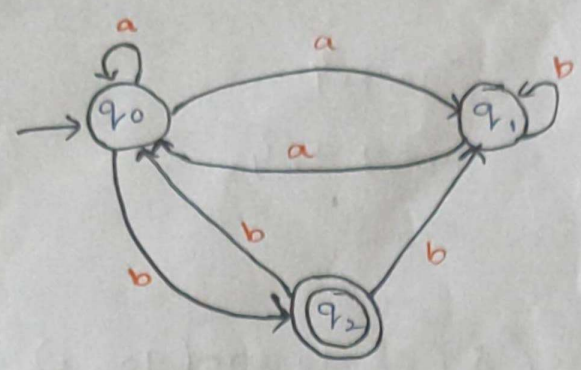
Code of Conduct:

I Om Sai Reddy 192372252 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Given state diagram:-



Convert the given NFA to DFA :-

Step 1 :-

Transition Table

state IP	a	b
q ₀	{q ₀ , q ₁ }	{q ₂ }
q ₁	{q ₀ }	{q ₁ }
q ₂	{ ϕ }	{q ₀ , q ₁ }

Step 2 :-

Formula = $\{Q, \Sigma, \delta, q_0, F\}$

$$Q = \{q_0, q_1, q_2\}$$

$$\Sigma = \{a, b\}$$

$$\delta(q_0, a) = \{q_0, q_1\}$$

$$\delta(q_2, a) = \phi$$

$$\delta(q_0, b) = \{q_2\}$$

$$\delta(q_2, b) = \{q_0, q_1\}$$

$$\delta(q_1, a) = \{q_0\}$$

$$\delta(q_1, b) = \{q_1\}$$

Step ③ :-

$$\epsilon\text{-closure}(q_0) = \{q_0, q_1, q_2\}$$

$$\epsilon\text{-closure}(q_1) = \{q_0, q_1\}$$

$$\epsilon\text{-closure}(q_2) = \{q_0, q_1, q_2\}$$

Step ④ :-

$$\begin{aligned} \delta'(q_0, a) &= \epsilon\text{-closure}(\delta(\epsilon\text{-closure}(q_0, a), a)) \\ &= \epsilon\text{-closure}(\delta(q_0, q_1, q_2), a) \\ &= \epsilon\text{-closure}(\delta(q_0, a) \cup \delta(q_1, a) \cup \delta(q_2, a)) \\ &= \epsilon\text{-closure}(\delta(q_0, q_1) \cup \{q_0\} \cup \{\emptyset\}) \\ &= \epsilon\text{-closure}(q_1 \cup q_0) \\ &= \epsilon\text{-closure}(q_0, q_1) \cup (q_0, q_1, q_2) \end{aligned}$$

$$\delta'(q_0, a) = \{q_0, q_1, q_2\}$$

$$\begin{aligned} \delta'(q_0, b) &= \epsilon\text{-closure}(\delta(\epsilon\text{-closure}(q_0), b)) \\ &= \epsilon\text{-closure}(\delta(q_0, q_1, q_2), b) \\ &= \epsilon\text{-closure}(\delta(q_0, b) \cup \delta(q_1, b) \cup \delta(q_2, b)) \\ &= \epsilon\text{-closure}(q_2 \cup q_1 \cup \{q_0, q_1\}) \\ &= \epsilon\text{-closure}(q_2 \cup q_1 \cup q_0) \end{aligned}$$

$$\delta'(q_0, b) = \{q_0, q_1, q_2\}$$

$$\begin{aligned} \delta'(q_1, a) &= \epsilon\text{-closure}(\delta(\epsilon\text{-closure}(q_1), a)) \\ &= \epsilon\text{-closure}(\delta(q_0, q_1), a) \\ &= \epsilon\text{-closure}(\delta(q_0, a) \cup \delta(q_1, a)) \\ &= \epsilon\text{-closure}(\{q_0, q_1\} \cup \{q_0\}) \end{aligned}$$

(2)

$$= \epsilon\text{-closure}(\{q_1, q_0\})$$

$$\delta'(q_1, a) = \{q_0, q_1, q_2\}$$

$$\begin{aligned}\delta'(q_1, b) &= \epsilon\text{-closure}(\delta(\epsilon\text{-closure}(q_1), b)) \\ &= \epsilon\text{-closure}(\delta(q_0, q_1), b)) \\ &= \epsilon\text{-closure}(\delta(q_0, b) \cup \delta(q_1, b)) \\ &= \epsilon\text{-closure}(q_2 \cup q_1)\end{aligned}$$

$$\delta'(q_1, b) = \{q_0, q_1, q_2\}$$

$$\begin{aligned}\delta'(q_2, a) &= \epsilon\text{-closure}(\delta(\epsilon\text{-closure}(q_2), a)) \\ &= \epsilon\text{-closure}(\delta(q_0, q_1, q_2), a)) \\ &= \epsilon\text{-closure}(\delta(q_0, a) \cup \delta(q_1, a) \cup \delta(q_2, a)) \\ &= \epsilon\text{-closure}(\{q_0, q_1\} \cup \{q_0\} \cup \emptyset) \\ &= \epsilon\text{-closure}(\{q_0 \cup q_1\})\end{aligned}$$

$$\delta'(q_2, a) = \{q_0, q_1, q_2\}$$

$$\begin{aligned}\delta'(q_2, b) &= \epsilon\text{-closure}(\delta(\epsilon\text{-closure}(q_2), b)) \\ &= \epsilon\text{-closure}(\delta(q_0, q_1, q_2), b)) \\ &= \epsilon\text{-closure}(\delta(q_0, b) \cup \delta(q_1, b) \cup \delta(q_2, b)) \\ &= \epsilon\text{-closure}(q_2 \cup q_1 \cup \{q_0, q_1\})\end{aligned}$$

$$\delta'(q_2, b) = \{q_0, q_1, q_2\}$$

$$\begin{aligned}\delta'(q_0, q_1, q_2, a) &= \epsilon\text{-closure}(\delta(\epsilon\text{-closure}(q_0, q_1, q_2), a)) \\ &= \epsilon\text{-closure}(\delta(q_0, a) \cup \delta(q_1, a) \cup \delta(q_2, a)) \\ &= \epsilon\text{-closure}(\{q_0, q_1\} \cup \{q_0\} \cup \emptyset)\end{aligned}$$

$$\delta'(q_0, q_1, q_2, a) = \{q_0, q_1, q_2\}$$

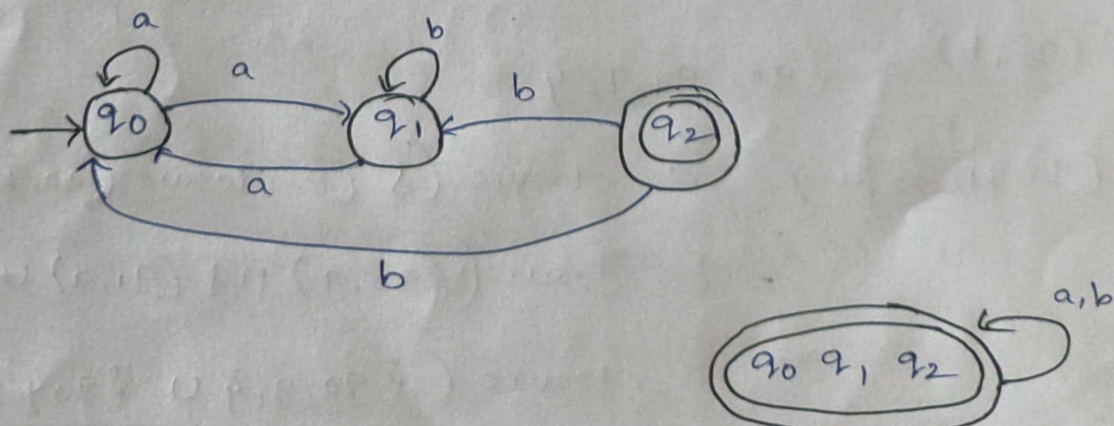
$$\begin{aligned}\delta'(q_0, q_1, q_2, b) &= \epsilon\text{-closure}(\delta(\epsilon\text{-closure}(q_0, q_1, q_2), b)) \\ &= \epsilon\text{-closure}(\delta(q_0, b) \cup \delta(q_1, b) \cup \delta(q_2, b)) \\ &= \epsilon\text{-closure}(q_2 \cup q_1 \cup \{q_0, q_1\})\end{aligned}$$

$$\delta'(q_0, q_1, q_2, b) = \{q_0, q_1, q_2\}$$

Transition Table :-

State IP	a	b
q_0	$\{q_0, q_1\}$	$\{q_2\}$
q_1	$\{q_0\}$	$\{q_1\}$
q_2	ϕ	$\{q_0, q_1\}$
q_0, q_1, q_2	$\{q_0, q_1, q_2\}$	$\{q_0, q_1, q_2\}$

NFA State diagram :-



(3)

Let L be the language of words of length atleast 2 whose last two letters are different from each other. For example $abaab$ is in L but not bb , Construct an NFA for L , then draw the equivalent DFA.

Language:-

Strings of length atleast 2 whose last two symbols are different.

Ex:-

Accepted
 $= * = * =$

- ① ab
- ② ba
- ③ $aaab$
- ④ $abaab$

Rejected
 $= * = * =$

- ① aa
- ② bb
- ③ a
- ④ b

NFA
 $* = *$

States

$q_0 = \text{start}$

$q_1 = \text{last Symbol } a$

$q_2 = \text{last Symbol } b$

$q_3 = \text{Final}$

Final State
 $= * = * =$

q_3

Transitions
 $= * = * =$

$q_0 \xrightarrow{a} q_1$

$q_0 \xrightarrow{b} q_2$

$q_1 \xrightarrow{a} q_1$

$q_1 \xrightarrow{b} q_3$

$q_2 \xrightarrow{b} q_2$

$q_2 \xrightarrow{a} q_3$

$q_3 \xrightarrow{a} q_1$

$q_3 \xrightarrow{b} q_2$

Equivalent DFA :-

S_0 = Start

S_1 = Ends with a

S_2 = Ends with b

S_3 = Ends with ab (Accept)

S_4 = Ends with ba (Accept).

Transition table :-

$=* =* =* =* =$

state	a	b
S_0	S_1	S_2
S_1	S_1	S_3
S_2	S_4	S_2
S_3	S_4	S_2
S_4	S_1	S_3

Final

State:-

S_3 S_4

③ Design an NFA for binary strings whose last two bits are the same.

A) Language :-

$=* =* =$

Ends with

00 (or) 11

Formal definition:-

$=* =* =* =* =$

$M = (Q, \Sigma, \delta, q_0, F)$

(4)

$$Q = \{q_0, q_1, q_2, q_3, q_4\}$$

$$\Sigma = \{0, 1\}$$

$$\text{Start} = q_0$$

$$\text{Final} = \{q_3, q_4\}$$

Transition Function:-

State	0	1
q_0	$\{q_0, q_1\}$	$\{q_0, q_2\}$
q_1	$\{q_3\}$	ϕ
q_2	ϕ	$\{q_4\}$
q_3	ϕ	ϕ
q_4	ϕ	ϕ

String:-

01011 ✓

$$q_0 \rightarrow 0$$

$$q_0, q_1 \rightarrow 1$$

$$q_0, q_2 \rightarrow 0$$

$$q_0, q_1 \rightarrow 1$$

$$q_0, q_2 \rightarrow 1$$

q_4

Since

q_4 is final.

$\therefore$ 01011 is Accepted.

(4) Construct DFA for strings containing 001 as a Substring?

Sol:- Language:-

$\epsilon^* \epsilon^* \epsilon^*$

$$\Sigma = \{0, 1\}$$

Need Substring:-

$\epsilon^* \epsilon^* \epsilon^*$

States:-
 $\Rightarrow * \Rightarrow$

q_0 = No match

q_1 = Seen 0

q_2 = Seen 00

q_3 = Seen 001 (Final)

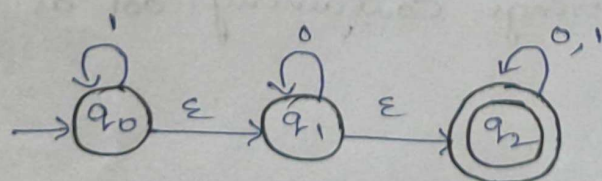
Transition table :-
 $\Rightarrow * \Rightarrow * \Rightarrow * \Rightarrow$

State	0	1
q_0	q_1	q_0
q_1	q_2	q_0
q_2	q_2	q_3
q_3	q_3	q_4

Final State:-

q_3 this DFA accepts every string containing 001.

- 5) A NFA with ϵ -transition is given below. Find ϵ -closure for each state and design an equivalent DFA which accepts the same language. Draw its transition table and mark the initial and final states.



A) Given:-

$q_0 \xrightarrow{\epsilon} q_1$

$q_1 \xrightarrow{\epsilon} q_2$

Final:-

$\Rightarrow * \Rightarrow$

q_2

5

q_1 loop on 0

q_2 loop on 0, 1

ϵ -closure :-
= * = * =

ϵ -closure (q_0) = $\{q_0, q_1, q_2\}$

ϵ -closure (q_1) = $\{q_1, q_2\}$

ϵ -closure (q_2) = $\{q_2\}$

DFA states :-
= * = * = *

$A = \{q_0, q_1, q_2\}$

$B = \{q_1, q_2\}$

$C = \{q_2\}$

Transition table :-

DFA state	0	1
A ←	B	A
B	C	C
C	C	C

Start state :-
= * = * = *

A

Final state :-
= * = * = *

94/100